THE TECHNICAL UNIVERSITY OF KENYA

FACULTY OF APPLIED SCIENCE & TECHNOLOGY

SCHOOL OF HEALTH AND BIOMEDICAL SCIENCES

DEPARTMENT OF HEALTH SYSTEMS MANAGEMENT AND PUBLIC HEALTH

UCCC 1102-05 DRUG AND SUBSTANCE ABUSE

LECTURE NOTES

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UCCC 1102-05: DRUG AND SUSTANCE ABUSE

Purpose of the course;

This is a comprehensive course that addresses the problems of drug abuse across the spectrum, including the latest thinking about prevention and awareness. This course provides an understanding of the origins and current status of drug use, treatment and prevention.

Expected Learning Outcomes

By the end of the unit the learner will be able to:

1. Understand the influence of drugs use, misuse and dependency.
2. Identify commonly abused drugs and substances.
3. Describe prevention and treatment strategies.
4. Explain the legal and moral codes regarding the prevention of drugs and substance abuse.

Course Content and Outline

Definition, abused drugs, marijuana, cocaine, tobacco, alcohol, hallucinogens, inhalants, opioid painkillers, signs and symptoms of drug abuse and addiction, control and prevention, effects of drugs on health, treatment of addiction and rehabilitation. Government policies on drugs and substance abuse.

Teaching Methods

Online Lecture.

Assessment Methods

Continuous Assessment Test:	30%
End of Semester Examination:	70%

Core Reading Materials for the Course

1. Mark A.R, Jonathan P. C, and Angela H (2014). Drugs and Drug Policy: What Everyone Needs to Know

2. Sarah A and David B (2017). Addiction, Behavioural Change and Social Identity: The Path to Resilience and Recovery. Paperback
3. Mark J, Gill V and Sarah L (2016). Alcohol, Drinking, Drunkenness: (Dis) orderly Spaces. Rout ledge(Paperback)

Recommended Reference Materials;

1. Torsten K, Betsy T and Geoffrey H (2016). The SAGE Handbook of Drug & Alcohol Studies: Social Science Approaches. SAGE
2. Daniel L and Daniel B (2016). A concise Guide to Alcohol and Drug Research: Implications for Treatment, Prevention and Policy

CHAPTER ONE: DRUGS, DRUG ADDICTION AND ITS CONSEQUENCES

Drug abuse is the use of illegal drugs or the use of prescription or over-the-counter medications in ways other than recommended or intended. It also includes intentional inhalation of household or industrial chemicals for their mind-altering effects. Tobacco use and problem drinking are sometimes included in the definition of drug abuse. Chemical abuse and substance abuse are terms sometimes used interchangeably with the term drug abuse, or they may be used to refer to a combination of drug abuse and tobacco use or problem drinking.

Addiction, or dependence on a particular substance or activity, is one of the most complex areas of mental health. Addiction can often be difficult to treat, and there is a good deal of controversy surrounding the causes of addiction and the best approaches to treatment. Individuals who find themselves experiencing an addiction to drugs or alcohol often find the services of a mental health professional to be helpful in overcoming the addiction.

People who abuse drugs may take them initially out of curiosity, to escape, to feel good, due to peer pressure, or for a variety of other reasons. Drugs can affect a number of different organs, and complications can result from damage to the brain or to other parts of the body. Other negative consequences often result from the effects drugs have on a person's mind, as well as actions an individual may take while under their influence. Treatment can be on an inpatient or outpatient basis, depending on the drug being abused, whether addiction is present, and whether there are coexisting health or psychological problems. Supervised withdrawal, also called detoxification (or detox), and may be necessary if physical symptoms are common when the drug is stopped. Medications may be used to decrease cravings, counter the effects of the drug, or to cause unpleasant reactions if the drug is used. Behavioral therapy is commonly an important part of treatment, providing skills, helping change attitudes and behaviors, and helping maintain recovery.

Drug abuse can have serious, even life-threatening, complications, such as drug overdose, alcohol poisoning, trauma, and suicidal or violent behavior. Seek immediate medical care for serious symptoms, including threatening, irrational or suicidal behavior; serious injury; respiratory or breathing problems; rapid, slow or absent pulse; chest pain or tightness; persistent vomiting; cold, clammy, or hot, dry skin; severe abdominal pain; seizure; or confusion or loss of consciousness for even a brief moment.

Signs that abuse has become addiction

Substance abuse can lead to substance dependence or addiction when both the amount of substance used and the rate of use increase. People who experience drug or alcohol addiction feel unable to control the impulse to use, and they often experience withdrawal symptoms in the sudden absence of the substance. Alcoholism, for example, occurs when people become chemically dependent on alcohol, and those who are addicted may become ill if they suddenly stop drinking. People may also feel psychologically dependent on a substance and continue to use it, particularly under stressful circumstances or to alleviate other psychological problems. Some people deny or are unaware that they have a problem with addiction, and sometimes a person's substance dependency and abuse remains hidden from loved ones.

Signs of chemical dependence include:

- Increasing tolerance, or the need to consume more of the substance to reach the desired altered state.
- Requiring the substance throughout the day.
- Seeking the company of other users and cutting off social ties with non-users.
- Dismissing or resenting expressions of concern from loved ones.
- Avoiding other activities and failing to meet obligations.
- Experiencing withdrawal symptoms in the absence of the substance.
- Hiding use from family and friends.
- Binging—using heavily—for many hours or several days.
- Feeling unable to quit.

What leads a person to misuse drugs or alcohol?

Psychological, biological, social, and physiological factors might all play a role in whether or not a person comes to abuse drugs or alcohol. A family history of substance abuse can make a person more vulnerable to addiction, and social factors, such as peer pressure and ease of availability can increase the likelihood of a person developing a problem with drugs or alcohol. In addition, once a person begins using heavily, physiological changes often take place, and that person may then become physically dependent, requiring him or her to continually use the substance in order to avoid withdrawal symptoms.

Alcoholism tends to run in families, although not all children of people addicted to alcohol become addicted themselves, and there is some debate among psychologists about the degree to which alcoholism is genetic. Some researchers are searching for an addiction or alcoholism gene, while others point out that simply witnessing a parent drink in response to stress increases a child's likelihood of choosing to drink in response to stress. Studies do show that genes, such as those that have an effect on the way a person responds to alcohol, may be responsible for about half of the risk of developing alcoholism. Some of these genes increase a person's risk, while some may instead decrease the risk of a person's developing alcoholism.

Research indicates that the vast majority of people who are addicted to drugs or alcohol have an underlying mental health condition or significant emotional/psychological difficulty. About half of all people with mental health diagnoses will face challenges with drugs or alcohol at some point in their lives, usually as a result of using drugs or alcohol to self-medicate.

People who misuse drugs often do so as a way of coping with experiences, memories, or events that emotionally overwhelm them. Whether they are equipped with appropriate coping strategies or not, people who misuse substances rely on the immediate gratification of drugs and alcohol as an alternative to facing the issues at hand. In the long term, however, reliance on drugs and alcohol will almost surely worsen any emotional or psychological condition. Chronic self-medication may be a sign that therapy is warranted in order to address an underlying condition or difficulty.

Warning signs of drug or alcohol abuse

Drug and alcohol abuse or misuse—excessive or inappropriate use of a substance—can be difficult to define, and people's opinions, values, and beliefs vary significantly on the topic. For some, any use of an illegal drug or any use of alcohol with the primary purpose of intoxication constitutes abuse. For others, abuse is indicated by recurring, negative consequences, such as:

- Failure to meet social, work, and academic obligations.
- Physical injury or illness.
- Alcohol- or drug-related legal problems, such as arrest for driving while intoxicated.
- Relationship problems with intimate partners, friends, and family.
- Impulsivity, such as spending money excessively.
- Diminished interest in other activities.
- Short-term memory loss or blackouts.

COMMONLY ABUSED DRUGS

Both legal and illegal drugs have chemicals that can change how your body and mind work. They can give you a pleasurable “high,” ease your stress, or help you avoid problems in your life.

Most of the abused drugs include: alcohol, marijuana/cannabis, khat/miraa, tobacco, cocaine, hallucinogens and opioids. The exact cause of substance abuse is not clear, with the two predominant theories being: either a genetic disposition which is learned from others, or a habit which if addiction develops, manifests itself as a chronic debilitating disease.

MARIJUANA/CANNABIS

People use cannabis by smoking, eating or inhaling a vaporized form of the drug. Cannabis often precedes or is used along with other substances, such as alcohol or illegal drugs, and is often the first drug tried.

Signs and symptoms of recent use can include:

- A sense of euphoria or feeling "high"
- A heightened sense of visual, auditory and taste perception
- Increased blood pressure and heart rate
- Red eyes
- Dry mouth
- Decreased coordination
- Difficulty concentrating or remembering
- Slowed reaction time
- Anxiety or paranoid thinking
- Cannabis odor on clothes or yellow fingertips
- Exaggerated cravings for certain foods at unusual times

Long-term (chronic) use is often associated with:

- Decreased mental sharpness
- Poor performance at school or at work
- Reduced number of friends and interests

What are the side effects of using marijuana?

Many people use marijuana over other drugs because it allows them to get high and feel like they can escape the stress of everyday life without the fear of overdose or death. While it is highly unlikely for someone to die from marijuana use, it does cause some side effects to your body and can lead to long-term health complications, including:

- Increased heart rate, breathing rate, and blood pressure.
- Dry mouth.
- Red eyes.
- Increased appetite.

Marijuana also affects the brain because the main ingredient, tetrahydrocannabinol (THC), enters the bloodstream and is carried to the brain where it reacts with several parts. It affects the pleasure-seeking part of the brain, which is why many users feel a high after using marijuana. This high causes euphoria, an intense feeling of happiness and excitement. Marijuana also affects the parts of the brain that control learning, memory, concentration, coordination, judgment, and time perception. This means that marijuana use can affect the brain in various ways and cause:

- Random thinking.
- Lack of perception of time.
- Memory loss.
- Inability to concentrate or learn new things.
- Impaired coordination.
- Difficulty with problem-solving.
- Paranoia.

Mental Health and Marijuana

Marijuana has also been linked to mental health problems. Typically, the younger the person is when they begin using marijuana, the more chance they have of developing a mental illness. Studies show that marijuana use, especially in high amounts, can increase rates of depression, schizophrenia and anxiety. However, it is also difficult to determine if the mental illness appears first and the person uses marijuana to self-medicate, or if marijuana causes the mental illness.

Physical Effects of Marijuana

Marijuana also affects the heart and lungs in certain ways. Marijuana increases one's heart rate, which can lead to palpitations or even a heart attack, especially within the first hour of use. Marijuana can affect the lungs because of the amount of carcinogens in marijuana smoke. In fact, it can contain up to 70 percent more carcinogens than tobacco smoke, and this can irritate the lungs. Marijuana can also increase the risk of lung infections and chest illnesses.

Marijuana also affects one's social life and career. According to studies, marijuana usage has many effects on one's career goals and achievements. Chronic marijuana use has been associated with higher rates of absences, accidents and job turnover, leading to decreased work performance and increased workers' compensation claims.

Risks of Marijuana Use

Long-term use of marijuana can cause serious effects to one's body. Here are some common physiological effects, according to

- Increased risk of respiratory problems, including lung cancer
- Decrease in sperm count and testosterone
- Impaired judgment and cognition
- Significant negative effects on social and career status

COCAINE

Cocaine is a powerfully addictive stimulant drug. The powdered hydrochloride salt form of cocaine can be snorted or dissolved in water and then injected. Crack is the street name given to the form of cocaine that has been processed to make a rock crystal, which, when heated, produces vapors that are smoked. The term "crack" refers to the crackling sound produced by the rock as it is heated.

How is Cocaine Abused?

Three routes of administration are commonly used for cocaine: snorting, injecting, and smoking. Snorting is the process of inhaling cocaine powder through the nose, where it is absorbed into the bloodstream through the nasal tissues. Injecting is the use of a needle to insert the drug directly into the bloodstream. Smoking involves inhaling cocaine vapor or smoke into the lungs, where

absorption into the bloodstream is as rapid as it is by injection. All three methods of cocaine abuse can lead to addiction and other severe health problems, including increasing the risk of contracting HIV/AIDS and other infectious diseases.

The intensity and duration of cocaine's effects—which include increased energy, reduced fatigue, and mental alertness—depend on the route of drug administration. The faster cocaine is absorbed into the bloodstream and delivered to the brain, the more intense the high. Injecting or smoking cocaine produces a quicker, stronger high than snorting. On the other hand, faster absorption usually means shorter duration of action: the high from snorting cocaine may last 15 to 30 minutes, but the high from smoking may last only 5 to 10 minutes. In order to sustain the high, a cocaine abuser has to administer the drug again. For this reason, cocaine is sometimes abused in binges—taken repeatedly within a relatively short period of time, at increasingly higher doses.

How Does Cocaine Affect the Brain?

Cocaine is a strong central nervous system stimulant that increases levels of dopamine, a brain chemical (or neurotransmitter) associated with pleasure and movement, in the brain's reward circuit. Certain brain cells, or neurons, use dopamine to communicate. Normally, dopamine is released by a neuron in response to a pleasurable signal (e.g., the smell of good food), and then recycled back into the cell that released it, thus shutting off the signal between neurons. Cocaine acts by preventing **Cocaine Information Sheet (continued),** the dopamine from being recycled, causing excessive amounts of the neurotransmitter to build up, amplifying the message to and response of the receiving neuron, and ultimately disrupting normal communication. It is this excess of dopamine that is responsible for cocaine's euphoric effects. With repeated use, cocaine can cause long-term changes in the brain's reward system and in other brain systems as well, which may eventually lead to addiction. With repeated use, tolerance to the cocaine high also often develops. Many cocaine abusers report that they seek but fail to achieve as much pleasure as they did from their first exposure. Some users will increase their dose in an attempt to intensify and prolong the euphoria, but this can also increase the risk of ad verse psychological or physiological effects.

What Adverse Effects Does Cocaine Have on Health?

Abusing cocaine has a variety of adverse effects on the body. For example, cocaine constricts blood vessels, dilates pupils, and increases body temperature, heart rate, and blood pressure. It can also cause headaches and gastrointestinal complications such as abdominal pain and nausea. Because cocaine tends to decrease appetite, chronic users can become malnourished as well.

Different methods of taking cocaine can produce different adverse effects. Regular intranasal use (snorting) of cocaine, for example, can lead to loss of the sense of smell; nosebleeds; problems with swallowing; hoarseness; and a chronically runny nose. Ingesting cocaine can cause severe bowel gangrene as a result of reduced blood flow. Injecting cocaine can bring about severe allergic reactions and increased risk for contracting HIV/AIDS and other blood-borne diseases. Binge-patterned cocaine use may lead to irritability, restlessness, and anxiety. Cocaine abusers can also experience severe paranoia—a temporary state of full-blown paranoid psychosis—in which they lose touch with reality and experience auditory hallucinations.

Regardless of the route or frequency of use, cocaine abusers can experience acute cardiovascular or cerebrovascular emergencies, such as a heart attack or stroke, which may cause sudden death. Cocaine-related deaths are often a result of cardiac arrest or seizure followed by respiratory arrest.

Added Danger: Coca ethylene

Poly drug use— use of more than one drug— is common among substance abusers. When people consume two or more psychoactive drugs together, such as cocaine and alcohol, they compound t danger each drug poses and unknowingly perform a complex chemical experiment within t bodies. Researchers have found that the human liver combines cocaine and alcohol to produce a third substance, coca ethylene, which intensifies cocaine's euphoric effects. Coca ethylene is associated with a greater risk of sudden death than cocaine alone.

What Treatment Options Exist?

Behavioral interventions—particularly, cognitive-behavioral therapy—have been shown to be effective for decreasing cocaine use and preventing relapse. Treatment must be tailored to the individual patient's needs in order to optimize outcomes—this often involves a combination of treatment, social supports, and other services.

Counseling and other types of therapy are the most common treatments for cocaine addiction. You may need to stay in a rehabilitation center (also known as rehab). Sessions with a trained therapist can help you make changes to your behaviors and thought processes. No medicines are approved to treat cocaine addiction.

The most important part of any treatment plan is that you give up the drug right away. Many people who are addicted to cocaine go through a phase called withdrawal when they first do this. Symptoms can include:

- Depression
- Anxiety
- Fatigue
- Trouble concentrating
- Increased hunger
- Cravings for the drug
- Nightmares
- Chills, nerve pain, muscle aches

A cocaine overdose is more difficult to treat. Physical signs include:

- Increased sweating, body temperature, or heart rate
- Trouble breathing
- Chest pain
- Nausea or vomiting
- Confusion, seizures, tremors

Watch for these mental signs of overdose, too:

- Anxiety
- Panic
- Paranoia
- Hallucinations
- Delirium

An overdose often leads to a stroke or heart attack.

TOBACCO

Tobacco is a plant grown for its leaves, which are dried and fermented before being put in tobacco products. Tobacco contains nicotine, an ingredient that can lead to addiction, which is why so many people who use tobacco find it difficult to quit. There are also many other potentially harmful chemicals found in tobacco or created by burning it.

How do people use tobacco?

People can smoke, chew, or sniff tobacco. Smoked tobacco products include cigarettes, cigars, bidis, and kreteks. Some people also smoke loose tobacco in a pipe or hookah (water pipe). Chewed tobacco products include chewing tobacco, snuff, dip, and snus; snuff can also be sniffed.

How does tobacco affect the brain?

The nicotine in any tobacco product readily absorbs into the blood when a person uses it. Upon entering the blood, nicotine immediately stimulates the adrenal glands to release the hormone epinephrine (adrenaline). Epinephrine stimulates the central nervous system and increases blood pressure, breathing, and heart rate. As with drugs such as cocaine and heroin, nicotine activates the brain's reward circuits and also increases levels of the chemical messenger dopamine, which reinforces rewarding behaviors. Studies suggest that other chemicals in tobacco smoke, such as acetaldehyde, may enhance nicotine's effects on the brain.

What are other health effects of tobacco use?

Although nicotine is addictive, most of the severe health effects of tobacco use comes from other chemicals. Tobacco smoking can lead to lung cancer, chronic bronchitis, and emphysema. It increases the risk of heart disease, which can lead to stroke or heart attack. Smoking has also been linked to other cancers, leukemia, cataracts, Type 2 Diabetes, and pneumonia. All of these risks apply to use of any smoked product, including hookah tobacco. Smokeless tobacco increases the risk of cancer, especially mouth cancers.

Electronic Cigarettes

Electronic cigarettes, also known as e-cigarettes or e-vaporizers, are battery-operated devices that deliver nicotine with flavorings and other chemicals to the lungs in vapor instead of smoke. E-

cigarette companies often advertise them as safer than traditional cigarettes because they don't burn tobacco. But researchers actually know little about the health risks of using these devices.

Pregnant women who smoke cigarettes run an increased risk of miscarriage, stillborn or premature infants, or infants with low birth weight. Smoking while pregnant may also be associated with learning and behavioral problems in exposed children.

People who stand or sit near others who smoke are exposed to secondhand smoke, either coming from the burning end of the tobacco product or exhaled by the person who is smoking. Secondhand smoke exposure can also lead to lung cancer and heart disease. It can cause health problems in both adults and children, such as coughing, phlegm, reduced lung function, pneumonia, and bronchitis. Children exposed to secondhand smoke are at an increased risk of ear infections, severe asthma, lung infections, and death from sudden infant death syndrome.

How can people get treatment for nicotine addiction?

Both behavioral treatments and medications can help people quit smoking, but the combination of medication with counseling is more effective than either alone.

Behavioral Treatments

Behavioral treatments use a variety of methods to help people quit smoking, ranging from self-help materials to counseling. These treatments teach people to recognize high-risk situations and develop strategies to deal with them. For example, people who hang out with others who smoke are more likely to smoke and less likely to quit.

Nicotine Replacement Therapies

Nicotine replacement therapies (NRTs) were the first medications the U.S. Food and Drug Administration (FDA) approved for use in smoking cessation therapy.

Current FDA-approved NRT products include chewing gum, transdermal patch, nasal sprays, inhalers, and lozenges. NRTs deliver a controlled dose of nicotine to relieve withdrawal symptoms while the person tries to quit.

Other Medications

Bupropion (Zyban®) and varenicline (Chantix®) are two FDA-approved non-nicotine medications that have helped people quit smoking. They target nicotine receptors in the brain, easing withdrawal symptoms and blocking the effects of nicotine if people start smoking again.

ALCOHOL ABUSE

Effects of Alcohol on the Body & Mind

Alcohol effects both the body & mind. Learn why there are physical and psychological consequences to over-drinking for both social drinkers or those diagnosed with alcohol use disorder.

Short-Term Effects of Alcohol

Although a person may not be abusing alcohol regularly, they can still experience its short-term effects on the mind and body. The liver can metabolize about one standard drink of alcohol per hour.[3] However, this can vary depending on a number of factors, including the individual's age, weight, liver function, and gender. Typically, consuming more than one beverage per hour can lead to intoxication, raising an individual's blood alcohol content (BAC) with each drink.[3]

The effects of alcohol can range from mild, such as skin flushing, to more severe symptoms such as passing out or vomiting.

Other short-term effects of alcohol can include:

- Lowered inhibitions, leading to poor social judgment.
- Trouble concentrating.
- Loss of coordination.
- Loss of critical judgment.
- Dulled perception, especially vision.
- Mood swings.
- Reduced core body temperature.
- Raised blood pressure.
- Passing out.
- Vomiting.

Long-Term Effects of Alcohol

Drinking too much over time can cause chronic physical and mental health issues. Heavy drinking can cause or contribute to liver damage, cardiovascular disease, and multiple types of cancer.

Long-term effects of excessive drinking may include:

- Diminished gray matter and white matter in the brain.
- Memory loss.
- Loss of attention span.
- Trouble learning.
- Alcoholic hepatitis.
- Liver fibrosis.
- Steatosis (i.e., fatty liver).
- Throat, mouth, larynx, breast, liver, colorectal, or esophageal cancer.
- High blood pressure
- Cardiomyopathy
- Stroke
- Irregular heart beat

Alcohol Poisoning

Binge drinking is a dangerous practice that can cause physical harm, it leads to a blood alcohol concentration (BAC) level of 0.08 g/dL and above. For adult women, that's typically around 4 drinks (5 for adult men) within a couple hours of each other.

Drinking too much can lead to alcohol poisoning.

Signs of alcohol poisoning include:

- Confusion.
- Nausea and vomiting.
- Slowed or irregular breathing.
- Cyanosis, or a blue-tinted skin.
- Pale skin.
- Low body temperature, or hypothermia.
- Unconsciousness.
- Seizures.

Effects of Alcohol Abuse on the Body

Chronic heavy drinking is associated with many serious health problems. Below are some of the ways alcohol may affect the body:

1. **Liver:** One of the possible severe medical consequences of chronic alcohol abuse is liver disease. Over time, with consistent alcohol abuse, the liver may become inflamed and/or scarred. Conditions such as fatty liver, alcoholic hepatitis, fibrosis, and cirrhosis may develop. A person may also develop liver cancer.
2. **Digestive system:** Alcohol can wear down the lining of the stomach and increase the production of stomach acid, which can contribute to ulcers. Alcohol may also alter nutrient breakdown, absorption, transportation, storage, and excretion, leading to nutrient deficiencies and/or trouble fully using nutrients. For example, thiamine deficiency is common and can lead to serious neurological issues. Alcohol can also impair blood sugar control.
3. **Pancreas:** Alcohol prompts pancreatic production of harmful substances, which can lead to pancreatitis. Pancreatitis is inflammation of the pancreas that impairs digestion.
4. **Central nervous system (CNS)***:* Thiamine, or vitamin B1, deficiency associated with chronic heavy drinking can lead to Wernicke–Korsakoff syndrome. Symptoms may include confusion, impaired coordination, learning problems, and memory difficulties. Liver disease can also harm the brain, resulting in symptoms such as sleep changes, alterations in mood, personality changes, depression, anxiety, impaired concentration, and incoordination. Too much alcohol may also hinder new brain cell growth.
5. **Cardiovascular (CV) health:** Drinking alcohol has complicated impacts on cardiovascular health. In 2016, alcohol-related CV diseases caused an estimated 593,000 deaths globally. Consuming too much alcohol is linked to high blood pressure, irregular heartbeat, trouble pumping blood through the body, blood clots, stroke, cardiomyopathy (sagging, stretched heart muscle), or heart attack. Excessive alcohol use, both directly and through malnutrition, can also lead to anemia.
6. **Reproductive health:** Consuming too much alcohol can lead to reproductive problems, including erectile dysfunction and irregular menstruation. Both men and women may have reduced fertility with long-term, heavy drinking. Women who drink while pregnant are at increased risk for miscarriage, stillbirth, or having a child with a fetal alcohol spectrum disorder (FASD).

7. **Bones:** Alcohol abuse can cause a calcium imbalance in the body, which is an important nutrient to maintain healthy bones. Consuming too much alcohol can also cause a disruption to the production of vitamin D, which is needed for calcium absorption. Lack of calcium increases the risk of osteoporosis. Osteoporosis increases the risk of bone fractures, which can cause serious pain and disability.

Psychological Effects of Alcohol

When it comes to the brain, alcohol acts as a depressant to the CNS. However, it can have inconsistent effects, exciting users under some conditions and sedating users under other conditions. Excitement, typically at lower doses, may be due to alcohol suppressing the inhibitory parts of the brain. Functions such as breathing, speech, thought, memory, and movement can be impacted by consuming alcohol. Mental effects may include mood changes, decreased inhibitions, relaxation, impaired judgment, slowed reaction times, difficulty remembering, confusion, and loss of consciousness. Chronic use of alcohol can lead to changes in the brain, as described in previous sections.

Excessive alcohol use, even if not chronic, can lead to alcohol–induced psychiatric syndromes, such as alcohol-induced depressive disorder, alcohol-induced bipolar disorder, alcohol-induced sleep disorder, alcohol-induced psychotic disorder, and more. These disorders are temporary and can occur after significant intoxication and/or withdrawal.

Alcohol use disorder is also linked to several mental illnesses which can develop separately from the disorder and may even predate it, such as major depression, some anxiety disorders, schizophrenia, bipolar disorder, and others. These disorders may increase the likelihood of alcohol-related issues, possibly due, in part, to using alcohol as a self-medicating substance. There also may be common underlying risk factors that increase the likelihood of both substance use disorders and mental illnesses.

HALLUCINOGENS

Use of hallucinogens can produce different signs and symptoms, depending on the drug. The most common hallucinogens are lysergic acid diethylamide (LSD) and phencyclidine (PCP).

LSD use may cause:

- Hallucinations

- Greatly reduced perception of reality, for example, interpreting input from one of your senses as another, such as hearing colors
- Impulsive behavior
- Rapid shifts in emotions
- Permanent mental changes in perception
- Rapid heart rate and high blood pressure
- Tremors
- Flashbacks, a re-experience of the hallucinations — even years later

PCP use may cause:

- A feeling of being separated from your body and surroundings
- Hallucinations
- Problems with coordination and movement
- Aggressive, possibly violent behavior
- Involuntary eye movements
- Lack of pain sensation
- Increase in blood pressure and heart rate
- Problems with thinking and memory
- Problems speaking
- Impaired judgment
- Intolerance to loud noise
- Sometimes seizures or coma

INHALANTS

Signs and symptoms of inhalant use vary, depending on the substance. Some commonly inhaled substances include glue, paint thinners, correction fluid, felt tip marker fluid, gasoline, cleaning fluids and household aerosol products. Due to the toxic nature of these substances, users may develop brain damage or sudden death.

Signs and symptoms of use can include:

- Possessing an inhalant substance without a reasonable explanation
- Brief euphoria or intoxication
- Decreased inhibition

- Combativeness or belligerence
- Dizziness
- Nausea or vomiting
- Involuntary eye movements
- Appearing intoxicated with slurred speech, slow movements and poor coordination
- Irregular heartbeats
- Tremors
- Lingering odor of inhalant material
- Rash around the nose and mouth

OPIOID PAINKILLERS

Opioids are narcotic, painkilling drugs produced from opium or made synthetically. This class of drugs includes, among others, heroin, morphine, codeine, methadone and oxycodone.

Sometimes called the "opioid epidemic," addiction to opioid prescription pain medications has reached an alarming rate across the United States. Some people who've been using opioids over a long period of time may need physician-prescribed temporary or long-term drug substitution during treatment.

Signs and symptoms of narcotic use and dependence can include:

- Reduced sense of pain
- Agitation, drowsiness or sedation
- Slurred speech
- Problems with attention and memory
- Constricted pupils
- Lack of awareness or inattention to surrounding people and things
- Problems with coordination
- Depression
- Confusion
- Constipation
- Runny nose or nose sores (if snorting drugs)
- Needle marks (if injecting drugs)

CHAPTER TWO: SIGNS AND SYMPTOMS OF DRUG ABUSE AND ADDICTION

Drug addiction, also called substance use disorder, is a disease that affects a person's brain and behavior and leads to an inability to control the use of a legal or illegal drug or medication. Substances such as alcohol, marijuana and nicotine also are considered drugs. When you're addicted, you may continue using the drug despite the harm it causes.

Drug addiction can start with experimental use of a recreational drug in social situations, and, for some people, the drug use becomes more frequent. For others, particularly with opioids, drug addiction begins with exposure to prescribed medications, or receiving medications from a friend or relative who has been prescribed the medication.

The risk of addiction and how fast you become addicted varies by drug. Some drugs, such as opioid painkillers, have a higher risk and cause addiction more quickly than others.

As time passes, you may need larger doses of the drug to get high. Soon you may need the drug just to feel good. As your drug use increases, you may find that it's increasingly difficult to go without the drug. Attempts to stop drug use may cause intense cravings and make you feel physically ill (withdrawal symptoms).

You may need help from your doctor, family, friends, support groups or an organized treatment program to overcome your drug addiction and stay drug-free.

What are the symptoms of drug abuse?

Symptoms of drug abuse include those of intoxication and those related to unfulfilled responsibilities and the social consequences of drug use.

Drug abuse can cause problems in interpersonal relationships, at home, on the job, and with the law. Symptoms of drug abuse related to these problems include:

- Feeling that you have to use the drug regularly — daily or even several times a day
- Having intense urges for the drug that block out any other thoughts
- Over time, needing more of the drug to get the same effect
- Taking larger amounts of the drug over a longer period of time than you intended
- Making certain that you maintain a supply of the drug
- Spending money on the drug, even though you can't afford it
- Not meeting obligations and work responsibilities, or cutting back on social or recreational activities because of drug use

- Continuing to use the drug, even though you know it's causing problems in your life or causing you physical or psychological harm
- Doing things to get the drug that you normally wouldn't do, such as stealing
- Driving or doing other risky activities when you're under the influence of the drug
- Spending a good deal of time getting the drug, using the drug or recovering from the effects of the drug
- Failing in your attempts to stop using the drug
- Experiencing withdrawal symptoms when you attempt to stop taking the drug

Recognizing unhealthy drug use in family members

Sometimes it's difficult to distinguish normal teenage moodiness or angst from signs of drug use. Possible indications that your teenager or other family member is using drugs include:

- **Problems at school or work** — frequently missing school or work, a sudden disinterest in school activities or work, or a drop in grades or work performance
- **Physical health issues** — lack of energy and motivation, weight loss or gain, or red eyes
- **Neglected appearance** — lack of interest in clothing, grooming or looks
- **Changes in behavior** — exaggerated efforts to bar family members from entering his or her room or being secretive about where he or she goes with friends; or drastic changes in behavior and in relationships with family and friends
- **Money issues** — sudden requests for money without a reasonable explanation; or your discovery that money is missing or has been stolen or that items have disappeared from your home, indicating maybe they're being sold to support drug use

Recognizing signs of drug use or intoxication

Signs and symptoms of drug use or intoxication may vary, depending on the type of drug. Below you'll find several examples.

- Balance problems, difficulty walking, and falls
- Change in mental status
- Changes in mood, personality or behavior
- Diminished reflexes
- Drowsiness or excessive energy

- Impaired balance and coordination
- Impaired judgment and memory
- Impaired vision
- Nausea with or without vomiting
- Pupil size changes
- Slurred speech; excessive talking

Changes in the brain

Physical addiction appears to occur when repeated use of a drug changes the way your brain feels pleasure. The addicting drug causes physical changes to some nerve cells (neurons) in your brain. Neurons use chemicals called neurotransmitters to communicate. These changes can remain long after you stop using the drug.

Risk factors

People of any age, sex or economic status can become addicted to a drug. Certain factors can affect the likelihood and speed of developing an addiction:

- **Family history of addiction.** Drug addiction is more common in some families and likely involves genetic predisposition. If you have a blood relative, such as a parent or sibling, with alcohol or drug addiction, you're at greater risk of developing a drug addiction.
- **Mental health disorder.** If you have a mental health disorder such as depression, attention-deficit/hyperactivity disorder (ADHD) or post-traumatic stress disorder, you're more likely to become addicted to drugs. Using drugs can become a way of coping with painful feelings, such as anxiety, depression and loneliness, and can make these problems even worse.
- **Peer pressure.** Peer pressure is a strong factor in starting to use and misuse drugs, particularly for young people.
- **Lack of family involvement.** Difficult family situations or lack of a bond with your parents or siblings may increase the risk of addiction, as can a lack of parental supervision.
- **Early use.** Using drugs at an early age can cause changes in the developing brain and increase the likelihood of progressing to drug addiction.
- **Taking a highly addictive drug.** Some drugs, such as stimulants, cocaine or opioid painkillers, may result in faster development of addiction than other drugs. Smoking or

injecting drugs can increase the potential for addiction. Taking drugs considered less addicting — so-called "light drugs" — can start you on a pathway of drug use and addiction.

Other life-changing complications

Dependence on drugs can create a number of dangerous and damaging complications, including:

- **Getting a communicable disease.** People who are addicted to a drug are more likely to get an infectious disease, such as HIV, either through unsafe sex or by sharing needles.
- **Other health problems.** Drug addiction can lead to a range of both short-term and long-term mental and physical health problems. These depend on what drug is taken.
- **Accidents.** People who are addicted to drugs are more likely to drive or do other dangerous activities while under the influence.
- **Suicide.** People who are addicted to drugs die by suicide more often than people who aren't addicted.
- **Family problems.** Behavioral changes may cause marital or family conflict and custody issues.
- **Work issues.** Drug use can cause declining performance at work, absenteeism and eventual loss of employment.
- **Problems at school.** Drug use can negatively affect academic performance and motivation to excel in school.
- **Legal issues.** Legal problems are common for drug users and can stem from buying or possessing illegal drugs, stealing to support the drug addiction, driving while under the influence of drugs or alcohol, or disputes over child custody.
- **Financial problems.** Spending money to support drug use takes away money from other needs, could lead to debt, and can lead to illegal or unethical behaviors.

CONTROL AND PREVENTION

The best way to prevent an addiction to a drug is not to take the drug at all. If your doctor prescribes a drug with the potential for addiction, use care when taking the drug and follow the instructions provided by your doctor.

Doctors should prescribe these medications at safe doses and amounts and monitor their use so that you're not given too great a dose or for too long a time. If you feel you need to take more than the prescribed dose of a medication, talk to your doctor.

Preventing drug misuse in children and teenagers

Take these steps to help prevent drug misuse in your children and teenagers:

- **Communicate.** Talk to your children about the risks of drug use and misuse.
- **Listen.** Be a good listener when your children talk about peer pressure, and be supportive of their efforts to resist it.
- **Set a good example.** Don't misuse alcohol or addictive drugs. Children of parents who misuse drugs are at greater risk of drug addiction.
- **Strengthen the bond.** Work on your relationship with your children. A strong, stable bond between you and your child will reduce your child's risk of using or misusing drugs.

Preventing a relapse

Once you've been addicted to a drug, you're at high risk of falling back into a pattern of addiction. If you do start using the drug, it's likely you'll lose control over its use again — even if you've had treatment and you haven't used the drug for some time.

- **Stick with your treatment plan.** Monitor your cravings. It may seem like you've recovered and you don't need to keep taking steps to stay drug-free. But your chances of staying drug-free will be much higher if you continue seeing your therapist or counselor, going to support group meetings and taking prescribed medication.
- **Avoid high-risk situations.** Don't go back to the neighborhood where you used to get your drugs. And stay away from your old drug crowd.
- **Get help immediately if you use the drug again.** If you start using the drug again, talk to your doctor, your mental health professional or someone else who can help you right away.

Diagnosis

Diagnosing drug addiction (substance use disorder) requires a thorough evaluation and often includes an assessment by a psychiatrist, a psychologist, or a licensed alcohol and drug counselor. Blood, urine or other lab tests are used to assess drug use, but they're not a diagnostic test for addiction. However, these tests may be used for monitoring treatment and recovery.

For diagnosis of a substance use disorder, most mental health professionals use criteria in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), published by the American Psychiatric Association.

TREATMENT OF ADDICTION AND REHABILITATION

Although there's no cure for drug addiction, treatment options explained below can help you overcome an addiction and stay drug-free. Your treatment depends on the drug used and any related medical or mental health disorders you may have. Long-term follow-up is important to prevent relapse.

Chemical dependence treatment programs

Treatment programs usually offer:

- Individual, group or family therapy sessions
- A focus on understanding the nature of addiction, becoming drug-free and preventing relapse
- Levels of care and settings that vary depending on your needs, such as outpatient, residential and inpatient programs

Detoxification

The goal of detoxification, also called "detox" or withdrawal therapy, is to enable you to stop taking the addicting drug as quickly and safely as possible. For some people, it may be safe to undergo withdrawal therapy on an outpatient basis. Others may need admission to a hospital or a residential treatment center.

Withdrawal from different categories of drugs — such as depressants, stimulants or opioids — produces different side effects and requires different approaches. Detox may involve gradually

reducing the dose of the drug or temporarily substituting other substances, such as methadone, buprenorphine, or a combination of buprenorphine and naloxone.

Opioid overdose

In an opioid overdose, naloxone, an opioid antagonist, can be given by emergency responders, or in some states, by anyone who witnesses an overdose. Naloxone temporarily reverses the effects of opioid drugs.

While naloxone has been on the market for years, delivery systems such as Narcan (a naloxone nasal spray) and Evzio (a naloxone injection device) are now available, though they can be very expensive.

Evzio is a small injection device that provides voice instructions to guide the user and automatically insert the needle into the thigh to deliver the naloxone injection. Whatever the method of delivery, seek immediate medical care after using naloxone.

Behavior therapy

As part of a drug treatment program, behavior therapy — a form of psychotherapy — can be done by a psychologist or psychiatrist, or you may receive counseling from a licensed alcohol and drug counselor. Therapy and counseling may be done with an individual, a family or a group. The therapist or counselor can:

- Help you develop ways to cope with your drug cravings
- Suggest strategies to avoid drugs and prevent relapse
- Offer suggestions on how to deal with a relapse if it occurs
- Talk about issues regarding your job, legal problems, and relationships with family and friends
- Include family members to help them develop better communication skills and be supportive
- Address other mental health conditions

Self-help groups

Many, though not all, self-help support groups use the 12-step model first developed by Alcoholics Anonymous. Self-help support groups, such as Narcotics Anonymous, help people who are addicted to drugs.

The self-help support group message is that addiction is a chronic disorder with a danger of relapse. Self-help support groups can decrease the sense of shame and isolation that can lead to relapse.

Your therapist or licensed counselor can help you locate a self-help support group. You may also find support groups in your community or on the internet.

Coping and support

Overcoming an addiction and staying drug-free require a persistent effort. Learning new coping skills and knowing where to find help are essential. Taking these actions can help:

- **See a licensed therapist or licensed drug and alcohol counselor.** Drug addiction is linked to a number of problems that may be helped with therapy or counseling, including other underlying mental health concerns or marriage or family problems. Seeing a psychiatrist, psychologist or licensed counselor may help you regain your peace of mind and mend your relationships.
- **Seek treatment for other mental health disorders.** People with other mental health problems, such as depression, are more likely to become addicted to drugs. Seek immediate treatment from a qualified mental health professional if you have any signs or symptoms of mental health problems.
- **Join a support group.** Support groups, such as Narcotics Anonymous or Alcoholics Anonymous, can be very effective in coping with addiction. Compassion, understanding and shared experiences can help you break your addiction and stay drug-free.

Preparing for your appointment

It may help to get an independent perspective from someone you trust and who knows you well. You can start by discussing your substance use with your primary doctor, or ask for a referral to a specialist in drug addiction, such as a licensed alcohol and drug counselor, or a psychiatrist or psychologist. Take a relative or friend along.

What you can do

Before your appointment, be prepared:

- **Be honest about your drug use.** When you engage in unhealthy drug use, it can be easy to downplay or underestimate how much you use and your level of addiction. To get an accurate idea of which treatment may help, be honest with your doctor or other mental health professional.
- **Make a list of all medications, vitamins, herbs or other supplements** that you're taking, and the dosages. Tell the doctor about any legal or illegal drugs you're using.
- **Make a list of questions to ask** your doctor or mental health professional.

Some questions to ask your doctor may include:

- What's the best approach to my drug addiction?
- Should I see a psychiatrist or other mental health professional?
- Will I need to go to the hospital or spend time as an inpatient or outpatient at a recovery clinic?
- What are the alternatives to the primary approach that you're suggesting?
- Are there any brochures or other printed material that I can have? What websites do you recommend?

What to expect from your doctor

Your doctor is likely to ask you a number of questions. Be ready to answer them to reserve time to go over any points you want to focus on. Your doctor may ask:

- What drugs do you use?
- When did your drug use first start?
- How often do you use drugs?
- When you take a drug, how much do you use?
- Do you ever feel that you might have a problem with drugs?
- Have you tried to quit on your own? What happened when you did?
- If you tried to quit, did you have withdrawal symptoms?
- Have any family members criticized your drug use?
- Are you ready to get the treatment needed for your drug addiction?

CHAPTER THREE: GOVERNMENT POLICY OF DRUG AND SUBSTANCE ABUSE DEMAND REDUCTION

Numerous efforts have been undertaken in order to reduce the demand for both licit and illicit drugs in the country especially to protect the live and welfare of youth. These include the following:

- Sustained public education and awareness including mass media campaigns, targeting youth in school and out of school
- Implementation of life skills programs mainly for the youth and vulnerable groups in institutions and amongst communities;
- Expansion and networking of treatment and rehabilitation facilities and other related services in the country including provision of protocols and standards for treatment of SUDs
- Holding of numerous stakeholder engagements targeting the Civil Society, Faith Based Organizations, Youth Organizations etc.at the National and County levels to build their capacity to manage and coordinate the campaign against ADA
- Focus on programmes that ensure the involvement of communities as the basic focus of a successful campaign against ADA
- Establishment and operationalization of public sector led, Inter-Agency collaboration in enforcement of alcohol and drugs related legislation through various committees at the National and County levels
- Research and surveillance by various public, private sector and civil society organizations e on new and emerging trends in drug abuse including the establishment of a National Drug Observatory in NACADA with the assistance of UNODC to provide a repository for all drug related surveys conducted;
- Institutionalization and mainstreaming of ADA in public and private sector due to increasing prevalence and development of a reporting system to monitor and evaluate trends.
- Provision of a Biannual Report on the status of ADA in the country which is regularly provided to Parliament. This is developed in partnership with National and County Government.

- The country has entrenched evidence-based prevention interventions on alcohol and drug abuse such as family-based, community based life skills, workplace based, school-based and media-based prevention in its programs
- Kenya has developed National Standards for Treatment and Rehabilitation facilities in line with international standards to ensure the provision of quality and
- holistic services to the public; Regular inspections are carried out to enforce the standards
- A 24-hour toll free call center, has been established to offer counseling and support services for persons with SUDs and their families

SUPPLY SUPPRESSION

The country has continued to develop, implement and review of policies and legislation on alcohol and drug control by various Ministries, Departments, Agencies as well as County Governments;

- Multi-agency efforts have been put in place to strengthen border control and transit points through provision of infrastructure, system and capacity building as well as strengthening surveillance and liaison with bordering countries to address the problem of trafficking and production and infiltration of counterfeit goods
- To ensure that precursor chemicals are utilized for the purposes intended, there has been increased enforcement of laws and regulations to control of harmful substances and precursor chemicals.
- The government has developed and operationalized standards to ensure safe and quality products through the various Ministries, Departments and Agencies that have a mandate to deal with ADA including measures to control production, marketing, trafficking and sale of both licit and illicit substance

The following are other achievements in the combat of alcohol and drug abuse;

- Enactment of several legislations to administer alcohol and drug abuse regime. They include the Narcotic Drugs and Psychotropic Substances (Control) Act, 1994.
- The government has designed and implemented mechanisms for improving existing working relations and effective engagement with CSOs including provision of funding for CSO programmes in alcohol control through the Alcoholic Drinks Control Act, 2010. The

Act is in the process of being reviewed to strengthen functions if liquor licensing and drug control which have been devolved to the County level.

- Instituted regular National Forums for sharing of experiences on ADA by the public sector institutions, civil society as well as local and international researchers
- Built a pool of trained of prevention and treatment professionals through establishment of a regular training program, in collaboration with international partners
- Inclusion of insurance coverage for addiction treatment and rehabilitation by the National Hospital Insurance Fund (NHIF).

CONTINUING AND EMERGING PROGRAMME CHALLENGES

The Government of Kenya took cognizance of alcohol and drug Abuse as a major threat to life and national development. Over the last decade, tremendous effort has been made in addressing ADA to ensure the country's social, economic and political stability necessary for the attainment of the Kenya Vision 2030 and SDGS.

Notable challenges are as follows:

- Poor access and unaffordability of Universal Health Care particularly by the poor and vulnerable members of society
- Lack of harmonized policies and legislation leading to disjointed efforts leading to policy gaps
- Weak enforcement of laws due to gaps in the criminal justice system such as unavailability of alternatives to incarcerations on ADA; flouting of standards and proliferation of licit and illicit drugs in the country, production and sale of counterfeit drugs, trafficking in various drugs
- Inadequate funding of alcohol and drug control programmes to ensure effective intervention
- Unaffordable and inadequately networked treatment and rehabilitation facilities
- Lack of data on drug mortality and morbidity to inform policy
- Emerging drug trends and attendant threats including new psychoactive substances, precursors and the non-medical use and misuse of pharmaceutical drugs
- Ease of availability and accessibility of substances of abuse, particularly by the youth and other vulnerable groups

- Complacence and tolerance in society towards use of harmful substances hence facilitating trafficking and infiltration of counterfeit drugs
- Non-accountability of industry players to take responsibility for the negative effects of drugs
- Poor enforcement of laws and standards, relating to adverts particularly on alcohol, hence the airing of adverts particularly during prime time
- Inadequate M&E and reporting structures and systems to determine impact of ADA at the National and County levels